

# Clinical development of cancer vaccines

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The limitations of current immune checkpoint inhibitor therapies highlight a need for improved strategies to expand tumor-reactive T cell repertoires in a way that is safe, selective and efficient. Cancer vaccines hold potential to achieve this goal, but the profound success of vaccines for infectious diseases has not yet translated to the cancer setting. Recently, however, encouraging preliminary results from phase 1 and 2 clinical trials have renewed enthusiasm and spurred the initiation of large-scale cancer vaccine trials. In this Review, we highlight insights from recent clinical trials, translational studies and preclinical models, which reveal critical factors for optimizing cancer vaccines. Key strategies include definition of improved proxies to estimate vaccine efficacy, selection of high-quality antigens—particularly neoantigens—using modular vaccine platforms with innate immunostimulatory capabilities, and a focus on early-stage cancer, as exemplified by (neo)adjuvant therapies. We discuss each of these in detail, outlining a roadmap for future cancer vaccine development.

The advent of immune checkpoint inhibitors (ICIs) has transformed the therapeutic landscape for multiple cancers, offering durable responses in subsets of patients<sup>1</sup>. ICIs partially restore T cell function, but the majority of patients fail to achieve long-term benefit. Moreover, the lack of intrinsic tumor selectivity in current immunotherapy strategies can lead to toxicities that offset clinical benefit, especially in cancers with low immunogenicity—underscoring the need for approaches that selectively expand and diversify tumor-reactive T cell repertoires.

Therapeutic cancer vaccines aim to induce or amplify such responses by delivering tumor-associated antigens (TAAs; potentially also expressed on nontumor cells) or tumor-specific antigens (TSAs; unique to tumor cells) in an immunostimulatory context. Historically, decades of cancer vaccine research yielded limited clinical success, attributed largely to T cell tolerance, inadequate innate immune activation, advanced disease stages at treatment initiation and lack of rational combinations with immunomodulatory agents<sup>2</sup>. However, these limitations are now being addressed through novel

antigen selection strategies emphasizing neoantigens and new classes of antigens coming from the ‘dark’ proteome (noncanonical, previously overlooked proteins), potent adjuvants and innovative delivery platforms that inherently provide both antigenic and immunostimulatory signals (Fig. 1). In addition, with novel vaccine platforms, production of clinical-grade, personalized cancer vaccines is now feasible in a time frame of about 2 months. Indeed, manufacturing aspects are an important component of cancer vaccine development, but a detailed discussion on this is beyond the scope of this Review.

Recent clinical studies in early-stage melanoma, pancreatic ductal adenocarcinoma, colon cancer and non-small-cell lung cancer (NSCLC) have demonstrated that neoantigen-targeted vaccines, particularly when combined with ICIs, can elicit broad, polyfunctional T cell responses associated with improved disease-free survival. These observations have reignited interest in therapeutic vaccination as a core component of precision immunotherapy. In this Review, we first outline the current landscape of cancer vaccines and the lessons from

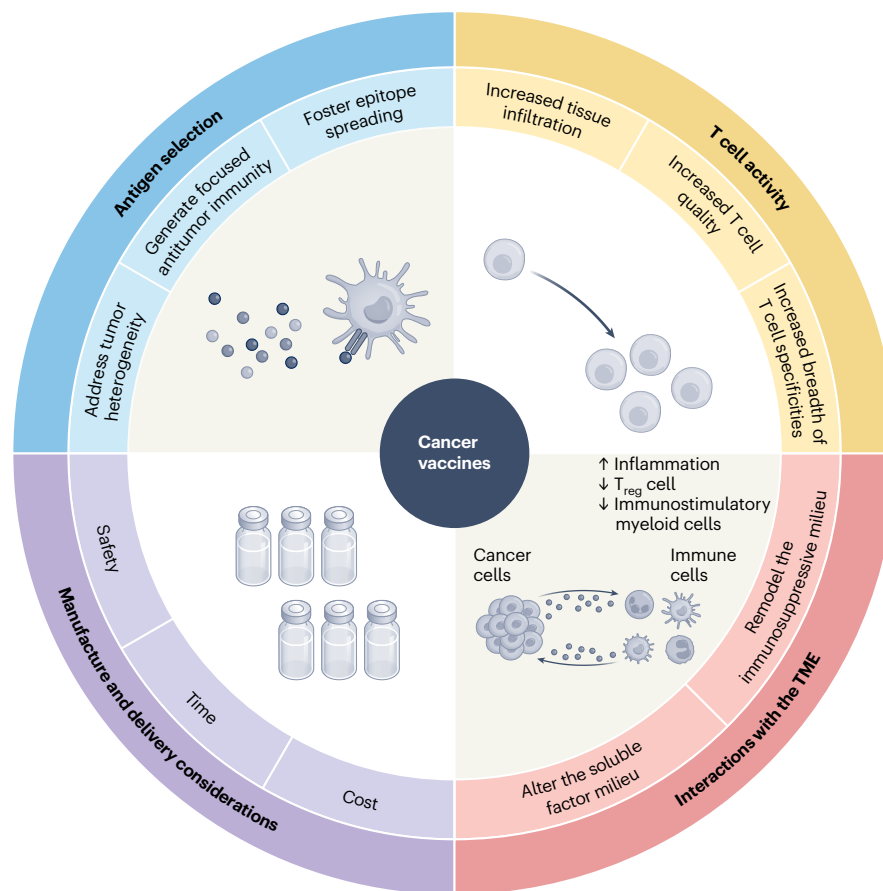
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**Fig. 1 | Key features of cancer vaccines.** This figure depicts important features for cancer vaccine development, including antigen selection, vaccine delivery, T cell activity and interaction with the TME, with increasing granularity going from the center to the outside. T<sub>reg</sub> cell, regulatory T cell.

past failures that are shaping future clinical studies. Then, we examine key immunological principles guiding vaccine design, emerging technologies and strategic integration with existing therapies to maximize clinical impact.

### Current clinical landscape of cancer vaccines

From a conceptual point of view, synergy may be expected between the boosting of a tumor-reactive T cell pool by vaccination and the use of ICI to maximize activity of the induced T cell pool. Furthermore, the regulatory approval of ICI for a series of malignancies has facilitated the clinical development of such combination therapies (Table 1). However, despite this strong biological rationale, history has shown that synergy between cancer vaccines and ICI cannot be assumed.

Specifically, an early randomized controlled trial compared a gp100 peptide vaccine, the ICI antibody ipilimumab (anti-CTLA-4), and their combination in chemotherapy-refractory advanced melanoma. This trial demonstrated an improved overall survival in both ipilimumab-containing arms relative to the gp100 vaccine arm, with no added benefit from combining ipilimumab with the vaccine<sup>3</sup>. Whether this lack of vaccine benefit was due to the nature of the antigen chosen, the vaccine format or something else, remains unresolved. Several vaccine platforms have likewise been combined with anti-CTLA-4 or anti-PD-1, aiming to improve clinical outcomes by inducing a strong tumor-specific T cell response. Although clinical and biological effects were observed, responses were either suboptimal or inconsistent across tumor types or patient populations<sup>4–7</sup>. Notably, an immune-modulating peptide vaccine targeting the IDO and PD-L1 TAAs combined with nivolumab showed an objective response rate exceeding 75%, including >50% complete remissions in anti-PD-1-naïve

metastatic melanoma<sup>8–10</sup>. However, in a pivotal phase 3 trial, the combination of adjuvanted imsaepimut (IDO) and etimupepimut (PD-L1) vaccines with pembrolizumab failed to achieve a statistically significant improvement in median progression-free survival compared to pembrolizumab monotherapy<sup>11</sup>.

An exception to these largely negative studies in late-stage disease comes from the phase 3 study that resulted in the approval of sipuleucel-T, an autologous dendritic cell (DC)-based therapeutic vaccine, for castration-resistant prostate cancer on the basis of an overall survival benefit<sup>12</sup>. However, there was no improvement in progression-free survival or objective tumor responses, and consistent prostate-specific antigen declines were not observed, raising questions about the interpretation and mechanistic basis of the observed survival advantage<sup>13</sup>.

What lessons can be learned from these studies? First, these examples demonstrate that induction of vaccine-specific immunity, which was convincingly shown in many of these trials, does not equal clinical efficacy. As further discussed below, this may reflect the fact that the induced T cell response does not have substantial tumoricidal potential, or that this response, as measured in peripheral blood, correlates poorly with intratumoral T cell activity. Alongside questions relating to optimal immunological correlates of clinical activity, a second issue concerns the fact that cancer vaccines have historically been tested mainly in patients with metastatic cancer. Patients with late-stage disease frequently exhibit tumor progression at anatomical sites of disease (such as liver and brain) with immunosuppressive tumor microenvironments (TMEs). In addition to intratumoral immunosuppression, patients with metastatic cancer can also show systemic immunosuppression, reflected by the fact that the same ICI regimen

**Table 1 | Key therapeutic cancer vaccine trials (2020–2025)**

| Vaccine/trial                                        | Cancer type/setting                          | Platform                                                           | Phase                 | Key result                                                                                      | NCT number, citation                   |
|------------------------------------------------------|----------------------------------------------|--------------------------------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|----------------------------------------|
| mRNA-4157/V940 + pembrolizumab (KEYNOTE-942)         | Resected high-risk melanoma (adjuvant)       | Personalized mRNA neoantigen vaccine                               | Phase 2b (randomized) | Improved recurrence-free survival versus pembrolizumab alone                                    | <a href="#">NCT03897881</a> (ref. 22)  |
| Autogene cevumeran (BNT122/RO7198457)                | Resected pancreatic ductal adenocarcinoma    | Personalized mRNA neoantigen vaccine                               | Phase 1               | Durable neoantigen-specific T cell responses; delayed recurrence                                | <a href="#">NCT04161755</a> (ref. 23)  |
| DCVax-L                                              | Glioblastoma (newly diagnosed and recurrent) | Autologous tumor-lysate DC vaccine                                 | Phase 3               | Extended overall survival versus historical controls                                            | <a href="#">NCT00045968</a> (ref. 136) |
| EVX-01 + pembrolizumab                               | Advanced melanoma                            | AI-selected neoantigen peptide vaccine                             | Phase 1/2             | Strong CD4 <sup>+</sup> CD8 <sup>+</sup> T cell responses; early clinical activity              | <a href="#">NCT05309421</a> (ref. 137) |
| BNT111 (FixVac) + cemiplimab                         | Advanced melanoma                            | Off-the-shelf mRNA vaccine (shared antigens)                       | Phase 2               | Improved ORR versus PD-1 alone in refractory patients                                           | <a href="#">NCT04526899</a> (ref. 138) |
| Personalized neoantigen peptide vaccine + nivolumab  | Newly diagnosed glioblastoma                 | Personalized neoantigen peptide vaccine                            | Phase 1               | Feasible with robust vaccine-induced T cell responses                                           | <a href="#">NCT02287428</a> (ref. 139) |
| Artificial adjuvant vector cells                     | Late-stage acute myeloid leukemia            | CD1d targeted HEK293 cells expressing WT1                          | Phase 1               | Doubled survival compared to multiple historical standard-of-care trials                        | NCT unknown <sup>130</sup>             |
| Personalized neoantigen peptide vaccine ± ipilimumab | Clear cell renal cell carcinoma              | Personalized neoantigen peptide vaccine including driver mutations | Phase 1               | Highly immunogenic (including drivers); no recurrence in 42 months                              | <a href="#">NCT02950766</a> (ref. 25)  |
| IDO/PD-L1 vaccination + nivolumab                    | First-line metastatic melanoma               | IDO and PD-L1 targeting peptide vaccine                            | Phase 1/2             | Median progression-free survival of 25.5 months and correlation with proteomic immune signature | <a href="#">NCT03047928</a> (ref. 10)  |

shows substantially greater toxicity in stage III melanoma than stage IV melanoma<sup>14,15</sup>. In addition, the increased genetic heterogeneity of metastatic tumors (for example, owing to prior chemotherapy or radiotherapy) and immune pressure through prior ICI, can increase the odds of antigen loss or impaired antigen presentation within tumors. Furthermore, prior therapies may reduce the pool of functional immune cells, further limiting vaccine efficacy. Finally, the very high tumor cell burden in advanced disease means a greater challenge to any vaccine, irrespective of other factors. Together, existing clinical studies highlight the challenges of achieving clinical benefit in advanced disease and call for a focus on early-stage or minimal residual disease (MRD) settings as the optimal context for the development of cancer vaccines.

### Vaccines in MRD

MRD, defined as the presence of microscopic cancer not detectable by imaging but measurable by circulating tumor DNA (ctDNA) or bone marrow assays, may define a favorable patient population for vaccination<sup>16</sup>. MRD has long been established in hematological malignancies, where it guides treatment decisions to delay relapse. More recently, MRD detection in solid tumors, predominantly via ctDNA, has emerged as a predictive biomarker for early disease recurrence<sup>17</sup>. Early clinical studies in pancreatic and colorectal cancer have demonstrated that neoantigen mRNA and peptide vaccines administered in the MRD setting can induce potent T cell responses and durable clinical remissions (exceeding 18 months) in some patients<sup>18,19</sup>. Larger randomized trials are ongoing to validate these observations and analysis of MRD conversion would appear an attractive surrogate endpoint in such studies.

### Vaccines during adjuvant or neoadjuvant immunotherapy

Patients receiving adjuvant ICI therapy in early-stage disease may benefit from incorporation of vaccines into treatment regimens. In resected stage III or IV melanoma, adjuvant anti-PD-1 therapy reduces relapse by approximately 20% compared to placebo, yet roughly 30% of patients still experience local or distant recurrence<sup>20,21</sup>. Personalized neoantigen mRNA vaccine platforms, guided by algorithms that

predict epitope presentation by human leukocyte antigen (HLA) class I and II molecules, have enabled the rapid generation of individualized vaccines. In the KEYNOTE-942 study, patients with resected stage III melanoma receiving adjuvant pembrolizumab plus a 34-neoantigen mRNA vaccine exhibited longer relapse-free survival and distant metastasis-free survival than those receiving single-agent adjuvant pembrolizumab<sup>22</sup>. Early translational analyses show that both ctDNA-positive and ctDNA-negative patients benefit from combination therapy over anti-PD-1 monotherapy, suggesting that the sensitivity of the ctDNA platform may not be sufficient to truly discriminate MRD from non-MRD. Similar studies in resectable pancreatic ductal adenocarcinoma demonstrate that vaccine-induced T cell responses correlate with improved relapse-free survival<sup>23,24</sup>. Moreover, patients with resected stage III and IV clear cell kidney cancer receiving adjuvant treatment with a personalized peptide vaccine either with or without ipilimumab did not show recurrence over a follow-up time of almost 3 years<sup>25</sup>.

Collectively, these trials provide early but intriguing data on the potential of the combination of cancer vaccines with adjuvant immunotherapy to improve outcomes in early-stage disease. A series of randomized controlled phase 2 and 3 trials (Table 2) have been initiated in these early disease patient populations to evaluate whether addition of personalized vaccines to standard of care improves outcome. The high expectations coming from this work are tangible in the research community, but still tempered by earlier disappointments.

An emerging question is whether, beyond their use in the adjuvant setting, cancer vaccines may also have utility in the neoadjuvant context. A broader shift toward neoadjuvant immunotherapy is currently ongoing, supported by compelling clinical evidence. Across an increasing number of early-stage cancers at high risk of recurrence, short-course preoperative ICIs have shown high rates of pathological complete response and favorable long-term outcomes<sup>26–28</sup>. A key open question is whether neoadjuvant vaccination can augment or prime T cell responses in conjunction with ICIs, and how the timing of vaccine administration relative to ICI initiation affects immune potency and clinical outcomes. Another crucial question is whether

**Table 2 | Ongoing therapeutic cancer vaccine trials**

| Vaccine/trial                                                 | Cancer type                                 | Platform                                | Phase/status       | NCT number                  |
|---------------------------------------------------------------|---------------------------------------------|-----------------------------------------|--------------------|-----------------------------|
| Autogene cevumeran (BNT122)                                   | Pancreatic ductal adenocarcinoma (adjuvant) | Personalized mRNA neoantigen            | Phase 2, ongoing   | <a href="#">NCT04161755</a> |
| GRT-C901/GRT-R902                                             | Metastatic colorectal cancer                | Personalized neoantigen prime–boost     | Phase 2/3, ongoing | <a href="#">NCT05141721</a> |
| Personalized neoantigen vaccine + radiation and pembrolizumab | Glioblastoma                                | Personalized neoantigen peptide vaccine | Phase 1            | <a href="#">NCT02287428</a> |
| INTerpath-001                                                 | Resected high-risk melanoma                 | Personalized mRNA neoantigen (V940)     | Phase 3, ongoing   | <a href="#">NCT05933577</a> |
| INTerpath-002                                                 | Resected NSCLC (stage II–IIIB)              | Personalized mRNA neoantigen (V940)     | Phase 3, ongoing   | <a href="#">NCT06077760</a> |
| INTerpath-009                                                 | Resected NSCLC post-neoadjuvant (non-pCR)   | Personalized mRNA neoantigen (V940)     | Phase 3, ongoing   | <a href="#">NCT06623422</a> |
| INTerpath-004                                                 | Renal cell carcinoma (adjuvant)             | Personalized mRNA neoantigen (V940)     | Phase 2, ongoing   | <a href="#">NCT06307431</a> |
| INTerpath-005                                                 | Muscle-invasive urothelial carcinoma        | Personalized mRNA neoantigen (V940)     | Phase 2, ongoing   | <a href="#">NCT06305767</a> |
| INTerpath-007                                                 | Cutaneous squamous cell carcinoma           | Personalized mRNA neoantigen (V940)     | Phase 2/3, ongoing | <a href="#">NCT06295809</a> |

pCR, pathological complete response.

vaccine-derived antigen makes a meaningful contribution to the induction of T cell responses in the neoadjuvant setting, in which tumor-derived antigens should still be abundant. Additionally, the possible occurrence of postoperative immunosuppression and the potential negative effect of removing tumor-draining lymph nodes that contain part of the tumor-reactive T cell pool. Although more data on these topics are required, we surmise that, in contrast to ICIs—for which neoadjuvant administration appears especially advantageous—cancer vaccines, either alone or in combination with ICIs, may be most optimally deployed in the adjuvant setting. Another potential approach could involve vaccination of only those patients who did not achieve a major or complete pathological response with neoadjuvant ICI-based therapy.

### Vaccines and cell therapy

Chimeric antigen receptor (CAR)-T cells are used as standard of care in the treatment of certain hematological malignancies, although no CAR-T cell product has received regulatory approval in solid cancers so far. The lower success rate of CAR-T cell products in solid cancers is partly explained by the scarcity of sufficiently safe and homogeneously expressed tumor antigens. In addition, the expansion and persistence of CAR-T cells that target solid tumor antigens is limited, providing a strong rationale for their combination with vaccines. In support of this, the combination of CAR-T cell therapy with a target antigen-encoding mRNA vaccine resulted in prolonged CAR-T cell persistence in mouse models<sup>29</sup>, and clinical evaluation of this approach is ongoing for the oncofetal antigen claudin-6 (ref. 30). The boosting of therapeutic cell activity by cancer vaccines is not restricted to CAR-T cell therapies; a study also demonstrated *in vivo* expansion of T cells with genetically engineered T cell antigen receptors (TCRs) when infused with a DC vaccine<sup>31</sup>. Finally, given the strong evidence that infused T cells in tumor-infiltrating lymphocyte (TIL) therapies recognize both TAAs and TSAs, strategic timing of cancer vaccination, either before TIL collection or after TIL infusion, could also be utilized to amplify the activity of TIL therapy.

### Proxies of vaccine potency

To further capitalize on the clinical opportunity of cancer vaccines, a deeper molecular and immunological understanding of what enables vaccines to improve cancer control is critical, starting with essential features of tumor-specific T cell responses. Given that clinical readouts of cancer vaccines can take years when such vaccines are explored in the (neo)adjuvant setting, what are the best metrics of vaccine potency that the field should utilize (Fig. 2)?

### $C_{\max}$ versus AUC

Vaccines may be characterized simply by their capacity to increase the number of tumor antigen-specific T cells. This purely quantitative view on vaccine-induced T cell immunity has dominated cancer vaccine development for many years, with peak T cell numbers (as measured by major histocompatibility complex (MHC) multimer staining, by ELISpot, or by intracellular cytokine staining) generally being used as a metric of success<sup>32</sup>. When viewing the induced T cell response as the ‘drug’ that is generated by vaccine administration, one could argue that peak T cell responses (that is,  $C_{\max}$ ) may not necessarily be the most important feature to analyze, and the area under the curve (AUC) of T cell response over time should perhaps be encouraged. Using AUC rather than  $C_{\max}$  would indicate the capacity of different vaccine formats to induce long-term T cell persistence, particularly important for extending relapse-free survival. At present, such an AUC analysis would require wet-lab testing of T cell responses over time. Should the field achieve the accurate *in silico* prediction of TCR–peptide–MHC reactivities<sup>33,34</sup>, longitudinal analysis of TCR repertoires may become a more scalable alternative.

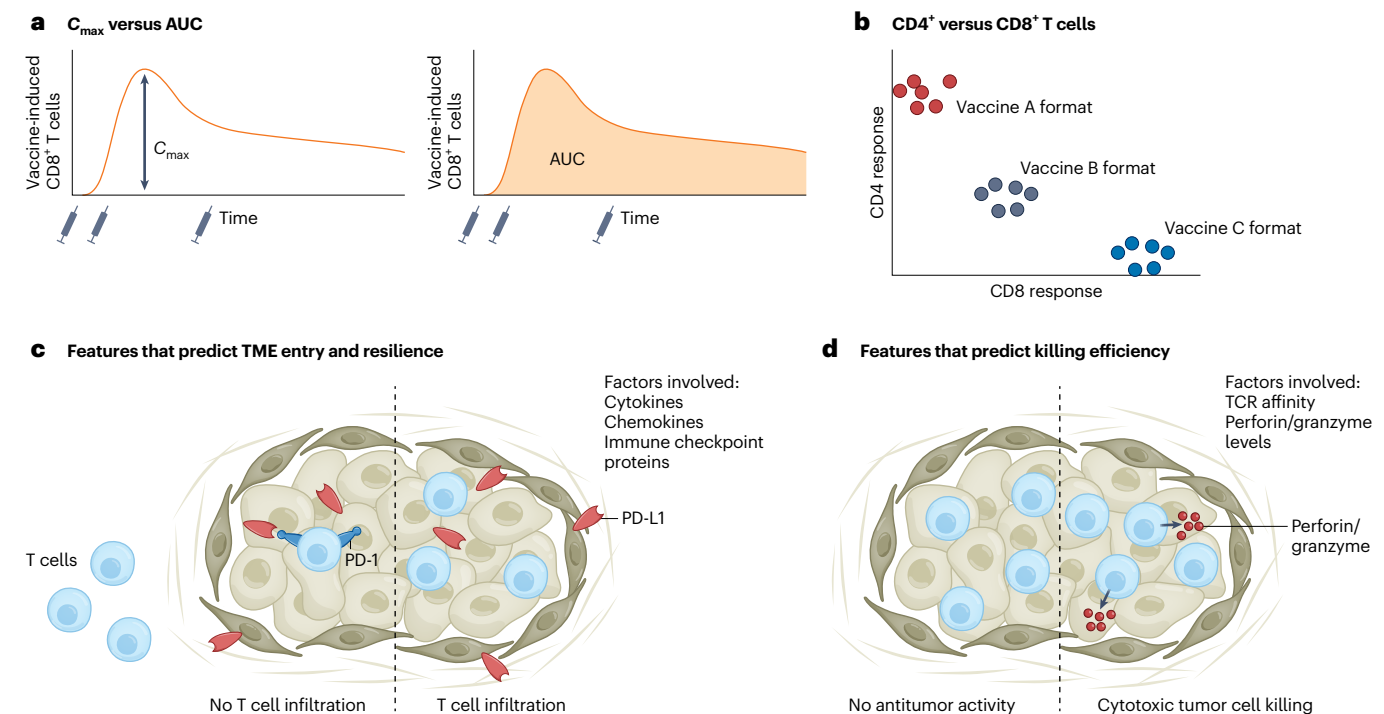
### Features that predict tumor entry and tumor cell killing

In addition to quantitative boosting of T cell responses, vaccines may be used to increase the quality of the tumor-specific T cell response. Relevant measures of T cell quality would include the capacity to enter the tumor site and ‘resilience’ or capacity to remain active within the TME. Importantly, as further discussed below, both properties are influenced by the conditions of T cell priming and hence may be influenced by vaccine design<sup>35</sup>. At present, data that inform on the capacity of clinically used vaccines to influence such T cell properties is relatively limited. Furthermore, while tumor entry of vaccine-induced T cells has been analyzed in serial biopsy samples, such analyses are not possible when cancer vaccines are developed as adjuvant therapies. Strategies to overcome these issues include identifying transcriptional or epigenetic features of T cells in peripheral blood that predict the capacity of these cells to enter the TME and to be active there. If such features can be identified in preclinical models or in patients with late-stage disease (for example, by comparison of cell states of peripheral blood T cell populations exhibiting low or high capacity to undergo activation at the tumor site), these features could be a valuable proxy in vaccine development in the adjuvant setting.

### Contribution of different T cell responses to tumor control

A metric of major importance, and one that has been so far understudied, is the potential of individual T cell responses to contribute to tumor





**Fig. 2 | Proxies of vaccine efficacy.** **a**, Vaccine potency is frequently assessed by examination of peak T cell responses ( $C_{\max}$ , left), whereas analysis of AUC (right) may better capture durability of vaccine-induced immunity. **b**, Vaccine formats (here depicted as **a–c**) can differ in relative potency in induction of CD4<sup>+</sup> versus CD8<sup>+</sup> T cell responses, while at present optimal ratio between the two is unknown. **c**, Vaccine formats may differ in their ability to program T cells that

cannot (left) or can (right) efficiently enter the TME and remain active at that site. Identification of markers such as epigenetic signatures in peripheral blood T cells that can predict these behaviors would be attractive. **d**, Ability of vaccine-induced T cells to efficiently induce tumor cell killing (right) or lack of such an ability (left) remains understudied.

regression. Even when ignoring factors such as antigen heterogeneity and loss, TCR–peptide–MHC pairs show a major variability in the efficiency with which they induce tumor cell killing, and part of the T cell responses that are detected in tumors or induced by vaccination may largely be irrelevant from an efficacy point of view. As an example, the MART-1-specific TCR that was used in an early TCR gene therapy trial showed limited clinical activity, without the expected on-target toxicity<sup>36</sup>, and also only modest *in vitro* activity against melanoma cells relative to other TCRs<sup>37</sup>—even though this TCR clone was highly prominent among TILs and in peripheral blood following TIL infusion<sup>37</sup>. Similarly, T cell reactivity against a human papilloma virus (HPV) E6 epitope is frequently observed in patients with HPV-associated cancers, even though tumor recognition potential of these T cells has been shown to be modest in multiple studies<sup>38,39</sup>. Whether the frequent presence of T cell reactivities with seemingly little tumor recognition capacity is due to differential efficiency in cross-presentation versus direct presentation of the relevant antigens, reduced tumor cell expression of the relevant antigens or HLA alleles over time, or expansion of certain T cell clones due to somatic mutations, remains to be established. Regardless of the mechanism(s), the contribution of different T cell reactivities to tumor control is likely to form a central determinant of vaccine efficacy. There is a need for assays that not only measure the presence of T cell responses, but also efficiently capture information on the tumoricidal capacity of T cells that target different antigens.

### CD4<sup>+</sup> versus CD8<sup>+</sup> T cell responses

Another key consideration is the relative attention that should be given to the induction of CD8<sup>+</sup> and CD4<sup>+</sup> T cell responses. This question is of particular relevance as certain vaccine formats have been shown to predominantly induce CD4<sup>+</sup> T cell responses, in some cases even when vaccines were deliberately biased toward predicted MHC class I-restricted epitopes and, therefore, expected to activate CD8<sup>+</sup> cells<sup>18,19</sup>.

By contrast, arenavirus-based vectors appear to be particularly suited to induce strong CD8<sup>+</sup> T cell responses<sup>40,41</sup>.

When considering prior work on the prognostic or predictive value of intratumoral CD8<sup>+</sup> T cell versus CD4<sup>+</sup> T cell infiltrates<sup>42,43</sup>, it seems logical to prioritize the induction of CD8<sup>+</sup> T cell responses. However, a large fraction of T cells in human tumors consists of bystander cells that lack tumor reactivity<sup>44,45</sup>, rendering quantification of overall CD8<sup>+</sup> and/or CD4<sup>+</sup> T cell infiltrates less informative. The availability of increasingly robust phenotypic and gene expression signatures of tumor-reactive CD8<sup>+</sup> T cells and CD4<sup>+</sup> T cells<sup>45–49</sup> should help to address this issue in the coming years, by distinguishing bystander from reactive cells. Looking at clinical interventions that selectively boosted tumor-reactive CD8<sup>+</sup> or CD4<sup>+</sup> T cell responses, a mixed picture emerges, with solid evidence for the clinical activity of CD8<sup>+</sup> T cell-enriched TIL and MHC class I-restricted TCRs, but also case reports describing the activity of CD4<sup>+</sup> T cell-biased cell products (reviewed in ref. 50). Arguably the most informative data points come from preclinical work, demonstrating how tumor antigen-specific CD4<sup>+</sup> T cell responses sculpt tumor antigen-specific CD8<sup>+</sup> T cell responses. Specifically, Ahrends et al.<sup>51</sup> showed how vaccine-induced CD4<sup>+</sup> T cells lowered immune checkpoint expression and enhanced migratory capacities of tumor-specific CD8<sup>+</sup> T cells. Earlier preclinical work demonstrated that helper T cell responses should preferably be directed against cognate tumor antigens rather than unrelated epitopes (for example, in the TME)<sup>52</sup>, consistent with a role of the induced CD4<sup>+</sup> T cell pool at the tumor site or in tumor-draining lymph nodes.

**De novo versus preexisting T cell responses.** A final key question in relation to proxies of vaccine efficacy is whether it is preferable to boost preexisting tumor-specific T cell responses or induce T cell reactivities that were previously undetectable, and arguments can be made to support both viewpoints. The boosting of preexisting T cell

responses ensures that a reactive TCR repertoire is present and that a sufficient level of *in vivo* antigen presentation may be possible, based on prior T cell activation. As argued above though, the mere presence of T cell responses should not be taken as definitive evidence for their strong tumoricidal potential.

Arguments for the induction of novel T cell reactivities are multifold. First, the resulting broadening of the tumor-specific T cell response may reduce the likelihood of tumor escape through antigen loss or HLA loss of heterozygosity, something that may be of particular relevance for tumors with a high degree of genetic heterogeneity. Second, it is unclear to what extent vaccines can reprogram T cells with a preexisting dysfunctional state in patients with cancer, or (perhaps more plausibly) exploit precursor or naive T cell pools with the same antigen specificities, to create a functionally competent T cell pool for these antigens. In future work, it will be important to compare the cell states of intratumoral T cells that have either been boosted by vaccination, or that were newly induced, to understand whether priming of the former under suboptimal conditions results in irreversible impairments. In the absence of such information, the choice between the boosting of preexisting T cell responses versus the induction of novel T cell responses may mostly be pragmatic: if strong evidence can be obtained (for example, through mass spectrometry) for tumor cell presentation of antigens for which no T cell response has yet developed, the induction of novel T cell reactivities may be attractive. On the other hand, for cases in which the probability that such putative antigens are truly presented by tumor cells at sufficient levels is low, it could be preferable to focus efforts on the enhancement of preexisting T cell reactivities.

## Optimal antigen selection

Induction of a high-quality and functional T cell response by vaccines is vitally dependent on the tumor antigen(s) that are incorporated. Hence, a key question is which tumor antigens are most immunogenic and which tumor antigens have the highest and most homogeneous expression on tumor cells. A defining feature of tumors is their extensive genomic, epigenomic, transcriptomic and translational heterogeneity<sup>53,54</sup>, and the implications, as well as the opportunities, arising from this heterogeneity are substantial.

Genetic diversity within most tumors argues for the targeting of multiple antigens<sup>55</sup>, including antigens derived from essential driver oncogenes (such as mutant *KRAS*, *TP53* or gene fusion events in pediatric tumors) where possible<sup>56–58</sup>. The optimal number of target antigens remains unknown and is unlikely to be precisely defined; therefore, a default strategy of maximizing antigen breadth, limited only by technical feasibility, seems preferable. Indirect support for broader targeting comes from a personalized pancreatic cancer vaccine study<sup>23</sup> in which the number of strongly immunogenic epitopes correlated with clinical outcome. On the other hand, if immune competition occurs between T cell responses, the inclusion of an increasing number of epitopes that show immunogenicity but that are poorly presented by the tumor cell mass could ultimately prove counterproductive.

## Tumor-associated and tumor-specific antigens

The selection of tumor antigens is guided by their capacity to initiate an effective and specific tumor-directed immune response, while also serving as measurable markers for monitoring the magnitude and quality of antitumor immunity. Since a pivotal study in 2013 (ref. 59), the field has dramatically shifted toward a preference for TSAs rather than TAAs.

TAAs have long been vetted as candidate antigen targets, largely discovered by differential gene or protein expression in tumor cells compared with normal tissue<sup>60</sup>. TAAs are generally expressed as a consequence of aberrant transcription factor activity or methylation patterns in tumor cells, resulting in heightened expression in cancer tissue. Within this class are the cancer/germline antigens (for example,

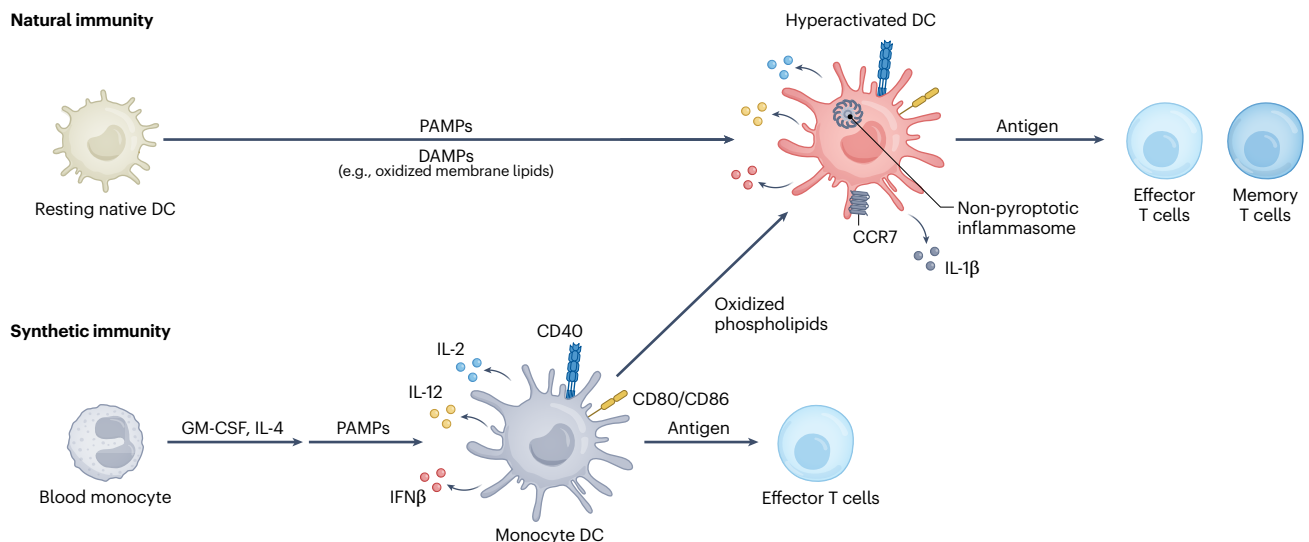
the MAGE family or NY-ESO-1 antigens), which are antigen targets that are present in germline tissues but silenced in mature differentiated tissues<sup>61–63</sup>. In addition, tissue differentiation antigens (for example, MART-1 and gp100 for melanocytes, PSA for prostatic cells) are categorized as TAAs<sup>64–67</sup>.

Among TSAs, the most widely studied class is formed by peptides encoded by cancer-specific somatic mutations, typically in canonical coding exons, also known as neoantigens. The key advantages of neoantigens are their exquisite tumor-specific expression and their high immunogenicity, as they arise from somatic events and hence are not subject to central tolerance<sup>59</sup>. The majority of mutations in many tumor types are ‘passenger’ mutations, and tend not to recur even among tumors of the same histology. In line with this, the T cell-recognized neoantigen landscape of human tumors is largely patient specific, a realization that spawned the growing field of ‘personal’ neoantigen vaccines<sup>68</sup>. Enabled by reliable short-read next-generation sequencing, early work on human cancer neoantigens largely focused on epitopes arising from single nucleotide variants and small insertions/deletions. More recently, advances in long-read sequencing and algorithmic approaches to extract information on structural variants, gene fusions and alternative splicing events, are expanding this important antigen class<sup>69,70</sup>.

The identification of neoantigens that arise from tumor-specific DNA mutations has been aided by prediction algorithms that estimate presentation of peptides by HLA class I and class II molecules<sup>71</sup>. Great strides have been made in growing the collection of computational tools to accurately predict HLA molecule binding by utilizing high-quality training data, particularly those arising from mass spectrometry-based immunopeptidomics, which enables direct physical detection of HLA-bound peptides on the tumor cell surface<sup>71</sup>. Until recently, immunopeptidomics was constrained by low detection sensitivity, but with improved instrumentation, better tissue processing approaches and improved analytics, sensitivity has dramatically increased. For example, hundreds of thousands of HLA allele-informed class I-binding peptides have been generated across more than 100 HLA alleles, enabling the generation of allele-specific and pan-allele prediction algorithms<sup>72,73</sup>.

Studies in tumors such as melanoma have provided evidence of tumor-resident T cells with reactivities against both TAAs and TSAs, confirming both antigen classes as immunogenic. In general, T cell avidity against TSAs is higher than against TAAs, which may be a consequence of thymic deletion of high-avidity T cells targeting (some of the) TAAs<sup>74</sup>. The observed relationship between tumor mutational burden and ICI response is likewise consistent with a major role for TSAs. However, a definitive analysis of the relative contribution of T cells specific for the two antigen classes to tumor control is lacking, and may also vary between tumor types.

**Noncanonical tumor antigens.** A growing area of investigation concerns the ‘dark’ or ‘noncanonical’ proteome, in which antigens beyond the canonical-annotated coding genome are being discovered. Such antigens may arise from novel peptide sequences that are generated by tumor-intrinsic mis-splicing events, or from cancer-specific expression and/or translation of noncanonical regions of the genome (so-called novel unannotated open reading frames)<sup>75–79</sup>. The events leading to expression of novel unannotated open reading frames are not fully understood, but include altered expression of transposable elements affecting nearby genes and epigenetic events leading to expression of endogenous retroviruses<sup>80,81</sup>. In addition, recent work has identified a class of noncanonical tumor antigens arising from aberrant RNA translation events that are induced by amino acid deprivation<sup>82–84</sup>. Therefore, translational rewiring in cancer can give rise to immunogenic peptides that are not predicted from genomic or transcriptomic analyses alone, further expanding the spectrum of potential tumor antigens. Provided that questions around tumor selectivity and homogeneity



**Fig. 3 | Nature versus nurture.** In general, currently applied adjuvants in conjunction with vaccines induce short-lived effector T cells. Novel insights in how to maximize DC activation (hyperactivated DC) using TLR agonists and oxidized phospholipids, lead to effective antigen presentation of vaccine

antigens, induction of tumoricidal effector T cells with tumor migratory capacity, and long-lived memory T cell formation required for lifelong T cell immunity. DAMPs, damage-associated molecular patterns; PAMPs, pathogen-associated molecular patterns.

of expression are sufficiently addressed, these novel broad classes of tumor antigens arising from the dark proteome could be particularly impactful for the large fraction of tumors that have a low mutational burden. Although currently, the phrase ‘dark proteome’ nicely captures the absence of sufficient information on this antigen class, future work should help researchers understand whether individual constituents are best placed in the class of TAAs or TSAs.

### Immunogenicity and antigenicity

Finally, a major current thrust is to determine whether predicted epitopes, or indeed even mass spectrometry-identified epitopes, are actually immunogenic. Particularly interesting approaches, although still to be fully vetted, include a linkage between immunogenicity and peptide hydrophobicity<sup>85</sup> and distinction from self<sup>86</sup>, and the use of *in vitro* validation tests<sup>87</sup>. A separate but critical question, given advancements in machine learning and structural predictions, is whether immunogenicity of epitopes can be predicted based on sequence or structural features<sup>33,34,88,89</sup>. Beyond direct empirical testing in engineered animal systems or with human cells, collecting a sufficiently large training dataset from which to infer the ‘rules’ of immunogenicity is an active, but still early, area of investigation. While immunogenicity of vaccine antigens is obviously an essential condition, it will be equally important to predict the per-cell killing potency of the induced T cell pools. The required understanding of antigenicity of different vaccine antigens will depend not only on the accurate prediction of epitope densities on tumor cells, but also on their integration with the functional avidity of the induced T cell pool.

### Recapitulating natural immunity

Successful cancer vaccine design will likely depend on integrating elements that recapitulate key aspects of naturally occurring, effective immune responses, such as those elicited by viral infection. DCs are the central orchestrators of antigen-specific T cell immunity against pathogens and also play a pivotal role in shaping the outcomes of cancer vaccination strategies<sup>90</sup>. The capacity of a vaccine to induce robust and durable T cell responses is fundamentally influenced by how DCs regulate the quality and magnitude of T cell activation. To optimally fulfill their immunostimulatory functions, DCs must engage in five key activities<sup>91</sup>: presentation of antigenic peptides on MHC class I and class II molecules to stimulate cognate TCRs; expression of costimulatory

molecules on the cell surface; secretion of effector T cell-stimulatory cytokines; secretion of cytokines that support memory T cell generation and maintenance; and upregulation of migratory capacity to access T cell-rich secondary lymphoid organs. Each of these five functions is essential for the induction of sustained antigen-specific T cell responses in the context of both infection and cancer, and should be harnessed for vaccine development.

### PRRs

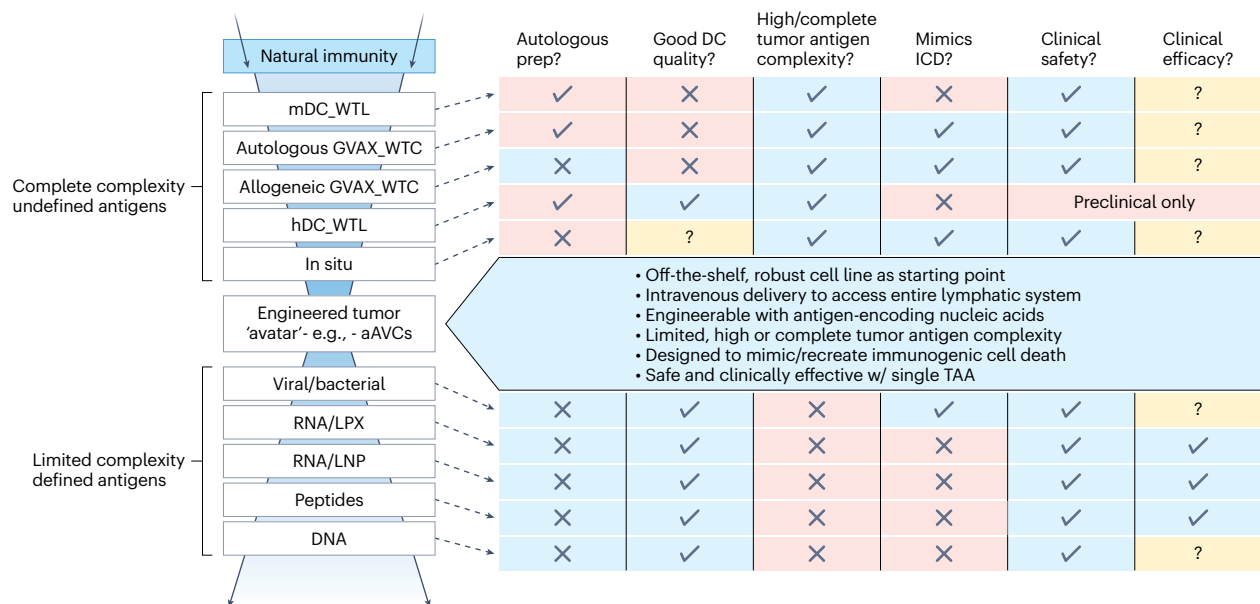
Recent efforts have focused on identifying molecular pathways within DCs that regulate T cell-stimulatory properties as potential therapeutic targets for vaccine development. Among these, innate immune pattern recognition receptors (PRRs) have attracted particular attention because of their capacity to enhance antigen uptake, processing and presentation, as well as their ability to induce inflammatory signaling cascades<sup>92,93</sup>. Toll-like receptors (TLRs) are the most extensively characterized PRRs, owing to their ability to promote peptide loading onto MHC molecules and to trigger the expression of genes encoding costimulatory molecules (CD40, CD80 and CD86) and effector cytokines such as interleukin-12 (IL-12), IL-2 and interferon- $\beta$  (IFN $\beta$ )<sup>94–100</sup>. Other PRR families—including RIG-I-like receptors, the cGAS–STING pathway and certain C-type lectin receptors—also contribute to these functions, albeit through distinct signaling mechanisms<sup>101–104</sup>.

The recognition that PRRs can induce multiple immunostimulatory pathways has spurred the development of PRR agonists as vaccine adjuvants. However, despite a strong immunological rationale and promising preclinical findings, clinical successes have been limited<sup>105–107</sup>. A plausible explanation is that no single PRR agonist can elicit all five DC activities required for full T cell activation. For example, TLR agonists and related PRR stimuli are generally poor inducers of bioactive IL-1 $\beta$  production—a cytokine crucial for memory T cell formation and reactivation<sup>108–110</sup>—and they fail to maximally enhance DC migration to lymph nodes<sup>110,111</sup>. These limitations may explain why strategies that focus on activating only a subset of DC functions (for example, CD40 agonists, recombinant IL-12 administration or single PRR agonists) have not achieved transformative clinical benefit<sup>105–107</sup>.

### Indicators of host integrity

Recent research in both animals and humans has begun to define strategies capable of simultaneously engaging all five DC functions.





**Fig. 4 | Vaccine delivery platforms.** Examined through the lens of natural immunity, each delivery format either has (✓) or does not have (✗) features that are either favorable (blue) or unfavorable (red) relative to optimally recapitulating natural immunity and being operationally feasible. Each has also shown uncertain (?) or more clear (✓) signals of clinical efficacy. A major conceptual dividing line is antigen complexity and definition, with vaccines based on whole cells providing a greater breadth of mostly unknown antigens while vaccines that rely on identified mutations (for example, selected by

prediction algorithms) providing a more limited number of well-defined antigens. Artificially creating an immunogenic tumor cell 'avatar'—a logistically facile cell engineered to undergo natural immune attack in lymphatic tissues following injection, and capable of expressing a few to many tumor antigens—may effectively bridge this gap. One example of a strategy currently under development involves artificial adjuvant vector cells (aAVCs<sup>129,130</sup>). ICD, immunogenic cell death; LNP, lipid nanoparticle; LPX, lipoplex; mDC, monocyte-derived DC; WTC, whole tumor cell; WTL, whole tumor lysate.

This concept draws on the seminal framework proposed by Charles Janeway Jr<sup>112</sup>, who hypothesized that the immune system detects microbial threats through evolutionarily conserved molecular patterns, leading to the activation of innate immune signaling and, ultimately, adaptive immunity. PRRs indeed recognize pathogen-associated molecular patterns such as lipopolysaccharides, driving DC activation and T cell priming<sup>113</sup>. However, this binary detection model—presence or absence of microbial components—does not fully capture the ability of DCs to gauge the severity of infection or tissue damage. Recent work introduces the concept of 'threat assessment', in which DCs integrate signals from both microbial molecules and indicators of host tissue integrity<sup>114</sup>. One such indicator is lipid oxidation, an ancient molecular signature of membrane damage<sup>114</sup>. When DCs simultaneously detect microbial products and oxidized lipids, they interpret this as a high-threat context, reflecting microbial invasion of damaged tissue, and undergo a distinct activation state termed hyperactivation<sup>114–116</sup>.

### Aiming for DC hyperactivating vaccines

Hyperactivated DCs exhibit an expanded functional repertoire beyond that of conventionally activated DCs. Specifically, they acquire the ability to robustly produce IL-1 $\beta$  via NLRP3 inflammasome activation and to upregulate the chemokine receptor CCR7, enhancing their migratory capacity to lymph nodes<sup>110,111</sup>. These features complement classical PRR-driven functions, effectively enabling all five DC activities required for robust T cell immunity. In preclinical studies, hyperactivated DCs demonstrated superior capacity to elicit CD4<sup>+</sup> and CD8<sup>+</sup> T cell responses that mediated tumor control across multiple implantable cancer models<sup>110,111</sup>. Notably, DC hyperactivation can also be achieved in human blood-derived DCs, suggesting potential translational relevance<sup>111</sup>. These findings raise the prospect that incorporating DC hyperactivators as adjuvants in cancer vaccines may overcome prior limitations and enhance clinical efficacy—a hypothesis now poised for rigorous clinical testing (Fig. 3).

### Vaccine delivery

A critical evaluation of cancer vaccine delivery platforms centers on two questions: first, which immune perturbations are required to achieve effective tumor cell killing, and second, whether current delivery strategies can reliably induce these states or are constrained by technical feasibility. Given the substantial remaining gaps in our understanding of both immunology and cancer biology, this section considers vaccine delivery through the lens of natural immunity.

The immune system detects stressed or dying pathogen-infected cells and initiates a cascade to identify novel antigens, promoting B cell and T cell responses that clear infection and prevent reoccurrence<sup>117</sup>. Translating this paradigm to cancer, the immune surveillance hypothesis and the concept of immunoediting, provides a conceptual framework for cancer vaccines<sup>118</sup>. These observations have informed a spectrum of approaches including: in situ vaccination, most closely resembling natural immunity; fully synthetic vaccines (peptides, proteins, mRNA or DNA), requiring selection of defined antigens; and semisynthetic vaccines, where natural biological vectors (DCs, bacteria or viruses) deliver defined or undefined antigenic material alongside natural or artificial adjuvants (Fig. 4). Each approach carries unique advantages, yet also limitations when compared to natural immune processes.

### In situ vaccines

These vaccines aim to (re)start an immunity cycle by delivering immune stimulators and/or inducers of immunogenic cell death to the TME<sup>119</sup>. Building on evidence from radiotherapy and oncolytic virus therapy, multiple modalities have been used<sup>119,120</sup>. Conceptually, a therapeutic window exists, enabled by central and peripheral tolerance, between stimulating antitumor immunity and provoking autoimmunity against critical self-antigens. Current embodiments of in situ vaccination appear safe and, in some contexts, may present fewer risks than ICIs, which<sup>121</sup> may exploit overlapping immune activation mechanisms. Operational challenges remain, however, including



tumor accessibility, generation of systemic immunity that extends beyond the injected lesion(s), and overcoming immunosuppressive networks within the TME.

### Synthetic vaccines

The success of mRNA vaccines against SARS-CoV-2 has reinvigorated interest in lipid nanoparticle-mediated delivery of nucleic acids<sup>122,123</sup>. However, prevention of infectious disease differs fundamentally from treatment of established malignancy, and effective tumor control depends primarily on robust T cell responses rather than antibody-mediated immunity. Early clinical studies in melanoma and pancreatic cancer have nevertheless demonstrated encouraging signals for mRNA-based cancer vaccines<sup>22,24</sup>. Similarly, synthetic long peptides have shown promising immunogenicity across multiple clinical trials<sup>19,124</sup>. Despite advances in high-throughput ‘omics’ technologies, however, the identification of optimal tumor antigens remains largely empirical. Moreover, although recent results are encouraging, most synthetic vaccine platforms only partially recapitulate natural immune processes. For example, the high local concentrations of synthetic peptides or artificial adjuvants administered subcutaneously or intradermally are not characteristic of natural physiological conditions, nor is intramuscular or systemic delivery of antigen-encoding nucleic acids, underscoring substantial opportunities for further optimization.

### Semisynthetic approaches

When natural biological components are exploited either as delivery vectors (such as engineered bacteria, viruses or DCs) or as sources of antigen (including whole tumor cells), these strategies seek to more comprehensively engage pathways of natural immunity. *Listeria*-based vaccines have been extensively investigated but raise concerns related to residual virulence<sup>125</sup>. Replication-incompetent adenoviral vectors<sup>126</sup> have been used to deliver a broad range of tumor antigens<sup>127</sup>, including neoantigens, although preexisting antiviral immunity can limit their effectiveness. Autologous, and potentially allogeneic, DC-based vaccines aim to exploit physiological mechanisms of immune priming<sup>128</sup>; however, DCs generated ex vivo often fail to fully recapitulate the functional and phenotypic properties of their in vivo counterparts. Moreover, bacterial, viral and ex vivo DC-based vaccine platforms typically require genetic encoding or peptide loading of predefined epitopes, which may restrict antigen breadth and limit confidence in the diversity and durability of the resulting T cell responses.

A promising vector-based strategy that closely aligns with natural immunity involves the use of antigen-bearing cells as delivery vehicles that are engineered to mimic immunogenic cell death, thereby creating functional ‘avatars’ of tumor cells outside the suppressive TME and subject to physiological immune processing. One such approach exploits lipid antigen presentation via CD1d to activate invariant natural killer (NK) T cells, initiating innate NK cell and NK T cell cytotoxicity followed by DC-dependent adaptive immune responses<sup>129</sup>. Fujii et al.<sup>130</sup> developed artificial antigen vector cells as tumor cell surrogates, using off-the-shelf cell lines engineered to overexpress CD1d and loaded with the lipid antigen  $\alpha$ -galactosylceramide together with selected tumor antigens. In patients with refractory acute myeloid leukemia, targeting Wilms’ tumor antigen with this platform was associated with a doubling of survival. These off-the-shelf cellular vectors enable systemic administration, recapitulate features of in situ vaccination outside the immunosuppressive tumor microenvironment, and can accommodate multiple antigens. Additional delivery modalities, including exosome-based approaches, remain comparatively underexplored<sup>131</sup>.

In contrast to these vector-based approaches, whole tumor cell vaccines enable presentation of the complete tumor antigen repertoire (including known and anonymous/uncharacterized antigens) through whole tumor cell loading of DCs<sup>132</sup>. This breadth of antigen coverage may overcome tumor heterogeneity and mimic natural immunity, favoring a ‘more is better’ paradigm. In vivo stimulation

of DCs targeting delivered antigens may enhance immunogenicity, as initially exemplified by granulocyte–macrophage colony-stimulating factor-secreting GVAX vaccines<sup>133,134</sup>. While granulocyte–macrophage colony-stimulating factor alone may be insufficient, alternative stimulators, including oxidized phospholipids or constitutively active cGAS, could better emulate hyperactivated DCs observed in natural immunity as discussed above. The logistics of isolating autologous tumor cells remain challenging, but molecular strategies to reproduce whole tumor cell proteomes from minimal tissue samples are under development<sup>135</sup>. In the meantime, until robust and durable tumoricidal T cell responses can be reliably induced, principles derived from natural immunity are likely to remain the most informative guide for cancer vaccine delivery design.

### Conclusion

In principle, we now possess the tools to design next-generation cancer vaccines capable of eliciting strong, durable and functional T cell responses—from both CD4<sup>+</sup> and CD8<sup>+</sup> compartments—against neoantigens and noncanonical antigens. These can be combined with potent DC-activating adjuvants to maximize innate immune stimulation and can be delivered through versatile platforms that allow rapid, scalable manufacturing.

As such ‘supervaccines’ emerge, their clinical efficacy will be shaped by the extent of systemic and intratumoral immunosuppression accompanying tumor progression, with the greatest potential likely to be seen in combination strategies involving ICIs and other immunomodulators, and with deployment of these in early-stage and MRD contexts.

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## Author contributions

All authors contributed to the writing of the manuscript, tables, and drawing of figures.

## Competing interests

J.B.H. is an advisor to AstraZeneca, BioNTech, BMS, Immatics, Iovance, Ipsen, Molecular Partners, Neogene Therapeutics, Novartis, Sastrā, T-Knife, Third Rock Ventures and Tzu; is a recipient of research grant support from Asher Bio, BioNTech, BMS and Sastrā; and is



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